

Pergunta 1

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Software is getting extremely pervasive in our life, and at the same time is getting harder to build with quality and efficiently. Since many years ago, the computing power had outpaced the ability of software engineers to effectively use those capabilities, resulting in many problems.

Which are in your opinion the problems that were never solved up until our days?

- a) Projects running over-budget, over-time, and unmanageable
- b) Software very inefficient and with low quality
- c) Software often not meeting requirements
- d) Software never delivered

c)

One problem that still needs solving is lack of communication, although there are many methodologies or processes that help to address this issue like user stories for example, it is still a prevalent one in current times, due to natural language being error inducing and up to interpretation..

Pergunta 2

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

A software process is a structured set of activities required to develop a software system.

Software processes (please justify your answer):

- a) Can be right or wrong
- b) Can be plan-driven, agile or a mix of the two
- c) Can be plan-driven or agile
- d) Involve very different process activities.

b)

A software process is plan-driven when all process activities are planned in advance and all progress is measured against that plan made beforehand.

In agile, the planning is incremental, which makes it easy to change the process according to need (any changes in the requirements). Most methodologies like Scrum and XP or RUP have characteristics of both of them, although some tend to lean more on one side.

Pergunta 3

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Which of these is not a good reason for keeping user stories small? Justify your answer.

- a) so that they are easier to estimate
- b) so that they can be delivered quickly
- c) so that only one team member is needed for each user story
- d) so that it's easy to test them and decide if they are really done

c)

User stories are a way of manifesting the requirements of a project. The team is not suppose to split up and each element work in a single requirement as that would not contribute to Collective Code Ownership (everyone is responsible for all code, making more pressure to produce great quality code) and that would also make pair programming very difficult (two programmers work in the same machine, one writing the code and the other paying attention to any error, this increases productivity as "two heads think better than one"). If this were to happen, the work produced would be of lesser quality.

Pergunta 4

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Model-view-controller is an architectural pattern which (justify your answer):

- a) Implies that the user request is received by the model
- b) Implies that the user request is received by the view
- c) Implies that a software framework (like Flutter or Ruby on Rails) is used
- d) Implies that user requests are received by the controller of a software framework

d)

On the MVC pattern the user sends a request that is received by the controller, who then requests the information from the model, getting a response back. The controller takes this response and sends the data to the View, which sends it to the user. This makes both a) and b) false. A software framework is not a language but a good architectural design practice, tried and tested in different environments, aka, a architectural pattern making option c) false.

Pergunta 5

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Requirements are often captured as User Stories. Who is responsible for writing the user stories? Provide a good example of a user story.

- a) The whole team;
- b) The product owner;
- c) The scrum master;
- d) The development team.

a)

Example: As a user, I can cancel a reservation on a website so I don't waste time calling the hotel.

It should include a who? (As a ...) a what? (I want to ...) and a why) (so that...)

Details can be added as conditions of satisfaction or smaller sub-stories

An example of a detail for this user story would be "As a premium site member, I can cancel a reservation until last minute."

Pergunta 6

Respondida

Pontuou 1,50 de 2,00

Destacar pergunta

eXtreme Programming is an iterative methodology that values simplicity and adaptability to change. Despite its advantages, it is not ideal for all software projects. Choose which of the following projects you believe XP would not be applicable to, justifying your choice.

- a) Designing a large corporation website;
- b) A project with a strict deadline and a specific set of features;
- c) A research project for a heart implant;
- d) A large scale micro-service cloud application.

c)

A research project usually requires a lot of planning beforehand by the team behind.

This makes it not adequate to XP. Not only does XP follow the approach of planning as you go (it doesn't have a specific planning phase at the beginning, for example) but it also leads to a lack of documentation.

This can be detrimental to the project (if someone wants to continue it in the future as is common on research projects, for example, that would be very hard as all they would have is unexplained code)

Pergunta 7

Respondida

Pontuou 1,00 de 2,00

Retirar destaque

Throughout the several development testing activities, testers progress from individual program units to testing the system as a whole. For a seamless progression, without major obstacles, along these phases, an important cross-cutting concern is:

- a) Keeping the same IDE throughout the several testing phases.
- b) Conform to the contracted interfaces.
- c) Automate whenever possible.
- d) Keep code coverage at a maximum.

c)

By automating the process as much as possible, not only do testers not have to worry about any human error occurring in the actual process of running the tests but it also makes the process a lot faster too.

Pergunta 8

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Open Source software is software that can be freely accessed, used, changed, and shared (in modified or unmodified form) by anyone. Open source software is made by many people, and distributed under licenses that comply with the Open Source Definition.

Which of the following statement is false?

- a) It is illegal to build and sell software that reuses open-source libraries;
- b) Open-source software can be used for personal profit purposes;
- c) A software is only open-source if distributed with an open-source license;
- d) Open-source licenses can be used to create commercial software.

a) Much sold software in current times is built using and reusing open-source libraries, much of that software is open-source too, leaving other developers with the chance to contribute to it if they wish to

Pergunta 9

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Briefly describe the most important phases and respective activities of RUP that you have used in your app project, explaining their purpose in the project and the corresponding results.

The most important phases of RUP are:

- Inception
- Elaboration
- Construction
- Transition

Phase 1 - Inception:

The purpose is to define the project scope and understand the problem. In this phase we analysed the problem at hand (making conferences better) and looked up previous projects to better understand how we could create something useful and feasible according to the curricular unit duration.

Phase 2 - Elaboration:

The purpose is to define the solution architecture. We analysed our project idea (app that sends notifications as conference timetable changes) and came to the conclusion that the best way to approach it would be through the MVC architectural pattern.

Phase 3 - Construction:

The purpose is to build the product. We implemented our app and tested accordingly through every iteration, making changes as we got feedback from the teacher

Phase 4 - Transition:

The purpose is to transition the product to end users. We didn't reach this phase as the final point in the project was a live demo. The goal would be to take feedback from that demo and better our project before transition phase.

Pergunta 10

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Briefly describe the most important agile practices that you have used in your app project, explaining their purpose in the project and the corresponding results.

We used Pair Programming ("two heads think better than one", two programmers are in the same machine (or a code live stream due to covid) and while one writes code the other critiques it, switching roles often), this makes the possibility of error much smaller and increases productivity;
Another practice was Collective Code Ownership (any developer can work in any part of the code), this makes the team be more aware of the whole project and not just the one person knowing what is happening in one section of the code, making it easier to debug, for example;
Lastly, another agile practice we used was Small Releases, starting with a small feature set, we released often from the beginning, in order to get more feedback so we could improve our project.